

California Housing Price Prediction Using Linear Regression

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Introduction

The California Housing Price Prediction project focuses on developing a machine learning model capable of estimating housing prices using the California Housing Dataset. Housing price prediction is an important application of data science because it helps buyers, sellers, investors, and policymakers make informed decisions. The dataset contains several demographic and geographical attributes that influence house values, making it suitable for predictive modeling.

The main objective of this project is to understand the complete machine learning workflow, including data collection, exploratory data analysis, preprocessing, model training, and performance evaluation. Linear Regression was selected as the baseline algorithm because it is simple, interpretable, and effective for understanding relationships between independent variables and a target variable. Through this project, practical knowledge of predictive analytics and machine learning techniques is developed.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to gain insights into the dataset and understand the characteristics of each feature. The California Housing Dataset contains variables such as median income, housing median age, total rooms, total bedrooms, population, households, latitude, and longitude. Statistical summaries were generated to identify the range, mean, standard deviation, and distribution of each variable.

Visualizations such as histograms, scatter plots, and correlation heatmaps were used to identify patterns and relationships among variables. The analysis revealed that median income has a strong positive correlation with median house value, indicating that regions with higher income levels generally have more expensive houses. Some features showed skewed distributions and potential outliers, which can affect model performance.

Model Building

Before building the model, the dataset underwent preprocessing steps to ensure data quality and consistency. Missing values were checked, and the dataset was divided into input features and the target variable. The target variable in this project is the median house value, while the remaining attributes serve as predictor variables.

The dataset was then split into training and testing sets using an 80:20 ratio. The training set was used to teach the Linear Regression model the relationship between housing features and house prices, while the testing set was used to evaluate its predictive ability on unseen data. The Linear Regression algorithm was implemented using the Scikit-learn library.

Evaluation Metrics

To measure the performance of the Linear Regression model, several evaluation metrics were used. These metrics help determine how accurately the model predicts housing prices and whether it generalizes well to new data.

Metric	Description	Example Value
MAE	Mean Absolute Error	0.53
RMSE	Root Mean Squared Error	0.74
R ²	Coefficient of Determination	0.58

MAE measures the average absolute difference between actual and predicted values. RMSE gives higher weight to larger errors, while the R² score indicates how much variance in housing prices is explained by the model.

Model Performance Analysis

The trained Linear Regression model produced reasonable predictions and successfully identified important relationships between housing attributes and house prices. Features such as median income and location-based variables contributed significantly to the prediction results.

Although the model performed well as a baseline approach, some prediction errors remained due to non-linear relationships within the dataset. This analysis highlights both the strengths and limitations of Linear Regression when applied to real-world housing data.

Improvement Ideas

Although Linear Regression provides a solid baseline model, there are several opportunities to improve prediction accuracy. Feature engineering techniques such as creating new variables, transforming skewed features, and selecting the most influential attributes can enhance model performance. Proper handling of outliers and missing values can also reduce prediction errors.

Additional improvements may include feature scaling, cross-validation, hyperparameter optimization, and the use of advanced machine learning algorithms such as Random Forest Regression, Gradient Boosting Regression, and XGBoost.

Conclusion

The California Housing Price Prediction project successfully demonstrates the use of Linear Regression for predictive modeling. The project covered the complete machine learning lifecycle, including data exploration, preprocessing, model training, and performance evaluation.

Overall, the Linear Regression model produced reasonable predictions and established a strong baseline for future improvements. This project helped develop practical skills in data analysis, machine learning, and model evaluation, which are essential for real-world data science applications.